

Alexander Belik

Software Engineering | AI Systems | Scientific Computing

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PROFILE

Theoretical physics graduate and current M.Sc. High-Performance Computing student at Trinity College Dublin. I build research platforms, GPU and HPC simulation tooling, machine-learning systems, and full-stack products across theoretical physics, scientific computing, and data-heavy software.

EDUCATION

M.Sc. High-Performance Computing

Trinity College Dublin, current

B.A. Theoretical Physics

Trinity College Dublin, 2022-2026

EXPERIENCE

Co-founder and Engineer

Aug 2026 - Present

Inhabis

- Building a local-first operating layer for connected homes: an embedded Matter controller with one persistent fabric per home, and an Android companion for BLE and Thread onboarding.
- Working on the planning and safety layers that check every action against household policy and verify the result, and on solar-surplus EV charging, solar forecasting and a learned room thermal predictor.

EuroHPC Student Ambassador

Aug 2026 - Sep 2026

EuroHPC Demo Lab 2026

- As a EuroHPC student ambassador, built fourteen GPU physics and AI demos and a walk-up viewer for the Demo Lab, rendered on Leonardo A100 nodes and Brain++ Discoverer GB200 nodes and replayed on a public stand.
- Wrote the SLURM job templates and run bundles that move saved runs between the clusters and a stand laptop.

Scientific Software Developer

May 2026 - Sep 2026

Trinity College Dublin

- Built a digital lab notebook platform storing experimental measurements as structured, auditable records.
- Developed GPU-accelerated physics demonstrations that run headlessly on the Leonardo EuroHPC system and on Brain++ Discoverer GB200 nodes, for the EuroHPC Demo Lab 2026.

Teaching Assistant

Oct 2025 - May 2026

Trinity College Dublin

- Led weekly tutorials and problem-solving sessions aligned with lectures.
- Graded assignments and exams with clear, constructive feedback.

Research Assistant

Jun 2025 - Sep 2025

Trinity College Dublin, CRANN Institute

- Designed Echo State Network reservoir models for quantum time-evolution data.
- Analysed model stability, convergence, and predictive accuracy across hyperparameters.

Software Engineer (EdTech)

Jun 2024 - Dec 2025

Grinds360

- Built Python automation tools for video and interactive lesson workflows.
- Contributed to UX iteration in Figma and full-stack product features.

Experimental / Computational Research Assistant

May 2024 - Sep 2024

Trinity College Dublin

- Created a Python spectroscopy control system for ferrofluids and ferronematic liquid crystals.
- Built automated acquisition and analysis workflows for repeated experimental data.

Exam Corrector

Jul 2021 - Jan 2024

The Dublin Academy of Education

- Corrected exams and coursework and supported marking workflows.

SELECTED WORK

HOME PLATFORM

Inhabis (in progress)

One system for the whole house: solar, heating, the car and every device, working inside rules the household sets.

- TypeScript server with separated layers for operational state, derived planning context, and constrained planning

Tools: TypeScript, Node.js, React, Python, Matter / Thread, Kotlin

Visit inhabis.ie

HPC AND GPU SIMULATION

EuroHPC Demo Lab

Fourteen physics and AI demos I rendered on Leonardo and Discoverer for the EuroHPC Demo Lab, played back on a public stand.

- Fourteen solvers spanning lattice-Boltzmann flow, particle-mesh cosmology, direct N-body gravity, black-hole lensing, reaction-diffusion, plasma control, molecular dynamics, and two evolving-AI games

Tools: Python, CuPy / CUDA, PyTorch, NumPy, SLURM, JavaScript

[View on GitHub](#)

DESKTOP WORKSPACE

Orion Mini (in progress)

A desktop app for my projects, tasks, reminders and transcribed recordings, with a local assistant built in.

- Multi-track transcription library with checkpoints, corrections, anchored notes, and timestamped exports

Tools: Python, PySide6 / Qt, React, TypeScript, CUDA, Whisper

RESEARCH DATA PLATFORM

LabFlow

A lab notebook for a materials research institute where every measurement is a structured, searchable record.

- React and TypeScript frontend over a FastAPI and PostgreSQL backend with versioned migrations

Tools: React, TypeScript, FastAPI, PostgreSQL, Docker, Tailwind CSS

TOOLKIT

Scientific computing: Python, NumPy, PyTorch, Mathematica, C / C++, Matplotlib, PySCF, Optuna

HPC and GPU: CUDA / CuPy, SLURM, CMake, Linux, Benchmarking, Parallel pipelines, N-body and lattice-Boltzmann solvers

Web and product: React, TypeScript, JavaScript, Node.js, Vite, Tailwind CSS, FastAPI, three.js, Playwright, Figma, Vercel

Home and mobile: Matter, Thread networking, BLE commissioning, Kotlin, Android, Jetpack Compose, Home energy and thermal models

Data and infrastructure: PostgreSQL, SQLite, Docker, Git / GitHub, GitHub Actions, Data pipelines

AI systems: Ollama, LangGraph, Local LLM workflows, Agent orchestration, Tool routing, Speech to text

Research practice: LaTeX, Quantum mechanics, Statistical mechanics, QFT, Semidefinite programming, Data analysis, Reproducibility